

Sydney Girls High School



MATHEMATICS EXTENSION 1

YEAR 11

Assessment Task 3

2018

Reading time : 5 minutes

Writing time : 60 minutes

Total marks : 60

Topics: Harder algebra and inequalities, Polynomials, Permutations and Combinations, Circle Geometry, Further Trigonometry, Parametric Parabola, Division of an Interval in a given ratio and Iterative methods.

General Instructions:

- There are four (4) questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- White-out or correction tape is NOT to be used.

Student Name:_____ **Teacher Name:**_____

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Question 1 (15 Marks)

- a) Solve $(x+3)(x-5) \leq 0$ and sketch the solution on a number line. 3
- b) Sketch $y = (x+5)(x-1)(3-x)$, showing all intercepts. 2
- c) Find the exact value of $\tan 75^\circ$. 2
- d) Solve $\frac{x}{x-2} \geq 3$. 3
- e) The lines $x-2y-5=0$, $x+3ky=15$ and $4y=3x+20$ are concurrent.
Find the value of k . 2
- f) Simplify $\frac{x^3-x^2}{8x^2+10x-3} \div \frac{4x^2-3x-1}{16x^2-1}$. 3

Question 2 (15 Marks)

- a) A curve has parametric equations $x = \frac{t}{2}$, $y = 3t^2$. Find the Cartesian equation for this curve. 2
- b) A committee of 6 people is to be formed from a group of 7 men and 5 women. Find the possible numbers of ways to form such a committee which contains :
- i) an equal number of men and women. 1
 - ii) at least 2 women. 2
- c) Given that $(x-3)$ and $(x+2)$ are factors of $x^3 - 6x^2 + px + q$, find the values of p and q . 3
- d) A chord of contact to the parabola $x^2 = 4y$, has the equation $y = x + 5$. Determine the external point from which the tangents are drawn. 2
- e) Given $t = \tan \frac{\theta}{2}$, show that $3\sin \theta + 4\cos \theta + 5 = \frac{(t+3)^2}{1+t^2}$. 2
- f)
- i) Express $3\sin \theta + 4\cos \theta$ in the form $R\sin(\theta + \alpha)$. 1
 - ii) Hence, solve $3\sin \theta + 4\cos \theta = 4$ where $0^\circ \leq \theta \leq 360^\circ$. 2

Question 3 (15 Marks)

a) α , β and γ are the roots of the equation $3x^3 - 4x + 6 = 0$. Evaluate :

i) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ 2

ii) $\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3$ 2

b) Prove that $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$, provided that $\cos 2\theta \neq -1$. 2

c) When a polynomial $P(x)$ is divided by $(x-3)$ the remainder is 5 and when divided by $(x-4)$ the remainder is 9. Find the remainder when $P(x)$ is divided by $(x-3)(x-4)$. 3

d) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. The equation of the chord PQ is $y = \left(\frac{p+q}{2}\right)x - apq$ (DO NOT PROVE THIS).

i) If PQ passes through the point $R(2a, 3a)$, show that $pq = p + q - 3$. 2

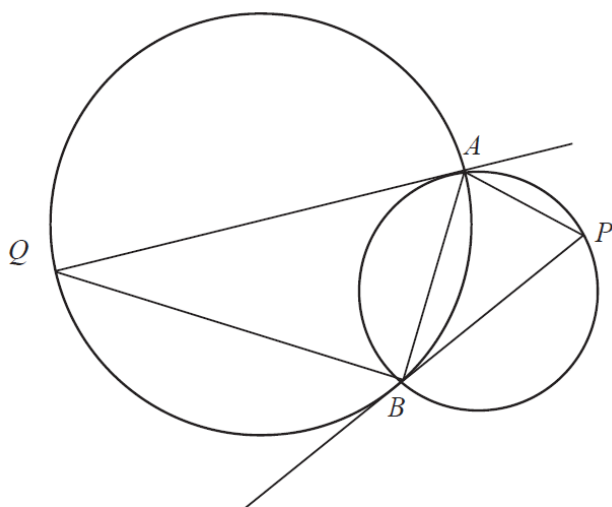
ii) If M is the midpoint of PQ , show that the coordinates of M are : 2
 $\left(a(pq+3), \frac{a}{2}((pq+3)^2 - 2pq)\right)$.

iii) Hence, find the equation of the locus of M . 2

Question 4 (15 Marks)

- a) At a dinner party Mr and Mrs Morrison and their six guests sit at a round table.
In how many ways can they be arranged if Mr and Mrs Morrison are to be separated? 2

- b) Two circles intersect at A and B , forming a common chord AB .
Tangents are drawn through A and B meeting the circles in Q and P respectively.

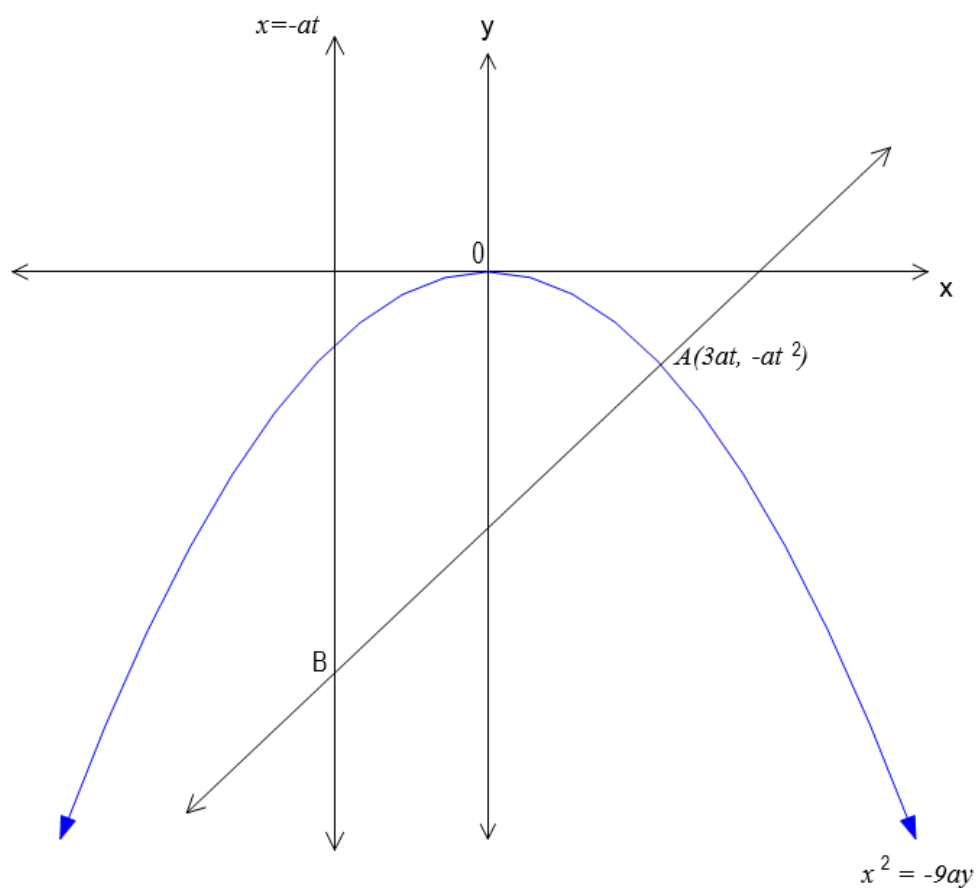


- i) Prove that triangle AQB and triangle PBA are similar. 2
- ii) Hence, or otherwise, prove $AB^2 = PA \times BQ$ 2
- c) The point $P(0, -2)$ divides the interval AB internally in the ratio $k : 1$.
If A is the point $(-3, 7)$ and B is the point $(1, a)$, find the values of a and k . 2

Question 4 (continued)

- d) The point $A(3at, -at^2)$ is a variable point on the parabola $x^2 = -9ay$.

The normal at A meets the line $x = -at$ at point B . Point C lies on the normal and divides interval AB externally in the ratio 2:3.

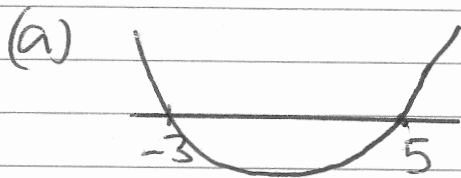


- i) Show that the equation of the normal to the parabola at A is
 $3x - 2ty = 2at^3 + 9at$. 2
- ii) Find the coordinates of B in terms of a and t . 1
- iii) Determine the coordinates of C in terms of a and t . 2
- iv) Show that the equation of the locus of C is a parabola and state the equation of its directrix in terms of a . 2

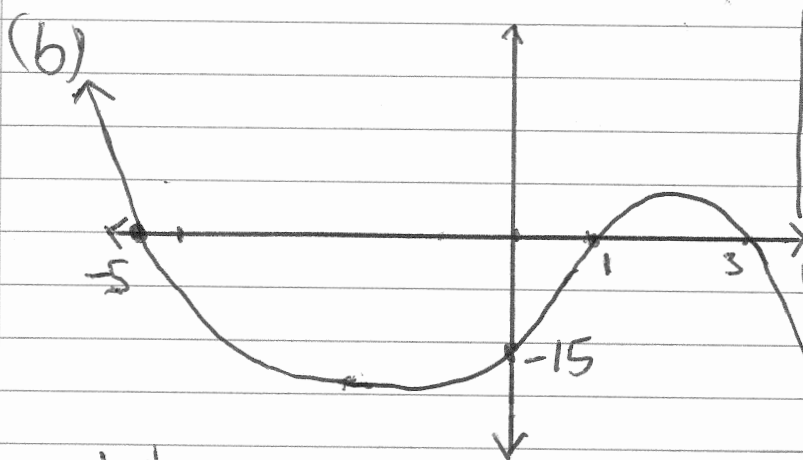
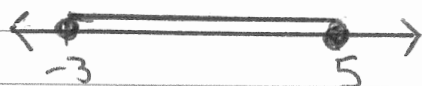
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YEAR 11 2018 EXT TASK 3

QUESTION ONE



$$-3 \leq x \leq 5$$



The question asked
for all intercept
that includes
the y-intercept

When $x=0$

$$y = (5)(-1)(3)$$

$$y = -15$$

(c) $\tan 75^\circ = \tan(30^\circ + 45^\circ)$

$$= \frac{\tan(30^\circ) + \tan(45^\circ)}{1 - \tan(30^\circ)\tan(45^\circ)}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}}$$

$$= \frac{\left(\frac{1}{\sqrt{3}} + 1\right)^2}{1 - \frac{1}{3}}$$

$$= \frac{\frac{1}{3} + \frac{2}{\sqrt{3}} + 1}{\frac{2}{3}}$$

$$= \frac{4 + \frac{6}{\sqrt{3}}}{2}$$

$$= 2 + \frac{3}{\sqrt{3}}$$

$$= 2 + \sqrt{3}$$

(d) $\frac{x}{x-2} \geq 3 \quad x \neq 2$

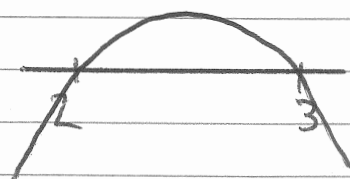
$$x(x-2) \geq 3(x-2)^2$$

$$(x-2)(x-3x+6) \geq 0$$

$$-2(x-2)(x-3) \geq 0$$

$$2 < x \leq 3$$

Many student
forget about
point where
 $x=2$



$$(e) \quad x - 2y - 5 = 0 \quad (A)$$

$$3x - 4y + 20 = 0. \quad (B)$$

$$2 \times (A) \quad 2x - 4y - 10 = 0$$

$$(B) - 2 \times (A) \quad x + 30 = 0$$

$$x = -30.$$

sub into (A)

$$-30 - 2y - 5 = 0.$$

$$-2y = 35$$

$$y = -\frac{35}{2}.$$

$$(-30, -\frac{35}{2}).$$

Sub into $x + 3ky = 15.$

$$-30 - \frac{105k}{2} = 15$$

$$-\frac{105k}{2} = 45.$$

$$-105k = 90.$$

$$k = -\frac{90}{105}$$

$$k = -\frac{6}{7}.$$

Concurrent:

means passing
through the
same point

$$(f) \frac{x^3 - x^2}{8x^2 + 10x - 3} \div \frac{4x^2 - 3x - 1}{16x^2 - 1}$$

$$= \frac{x^2(x-1)}{(2x+3)(4x-1)} \times \frac{(4x-1)(4x+1)}{(x-1)(4x+1)}$$

$$= \frac{x^2}{2x+3}$$

2 a) $k = 2x$

$$y = 3 \times (2x)^2 \\ = 12x^2$$

k is not the gradient of a tangent in this case.

b) i) ${}^7C_3 \times {}^5C_3 = 350$

ii) ${}^7C_4 \times {}^5C_2 + 350 + {}^7C_2 \times {}^5C_4 + {}^7C_1 \times {}^5C_8 = 812$

c) $P(3) = 0$

$P(-2) = 0$

$$3^3 - 6 \times 3^2 + p \times 3 + q = 0 \quad (-2)^3 - 6 \times (-2)^2 + p \times -2 + q = 0$$

$$-27 + 3p + q = 0$$

$$-8 - 24 - 2p + q = 0$$

$$3p + q = 27$$

$$-2p + q = 32$$

$$5p = -5$$

$$3x - 1 + q = 27$$

$$p = -1$$

$$q = 30$$

Those who used division of polynomials were not very successful.

d) chord of contact $x, y = 2a(y + y_1)$

$$4a = 4$$

$$a = 1$$

$$x = y - 5$$

$$2x = 2(y - 5)$$

$$\therefore \text{Lt} = (2, -5)$$

Trying to find the pt of intersection of the chord and the equations of the tangents is far more difficult.

e) $3 \times \frac{2t}{1+t^2} + 4 \times \frac{1-t^2}{1+t^2} + 5$

$$= \frac{6t + 4 - 4t^2 + 5 + 5t^2}{1+t^2}$$

$$= \frac{t^2 + 6t + 9}{1+t^2}$$

$$= \frac{(t+3)^2}{1+t^2}$$

f) i) $R = \sqrt{3^2 + 4^2} = 5$

$$\tan \theta = \frac{4}{3}$$

$$\therefore \theta = \frac{53.1^\circ + 71.4^\circ + 8.3^\circ}{2} = 34.5^\circ$$

$$\therefore 5 \sin(\theta + 53^\circ)$$

ii) $5 \sin(\theta + 53^\circ) = 4$

$$\sin(\theta + 53^\circ) = \frac{4}{5}$$

$$\theta + 53^\circ = 53^\circ, 127^\circ, 453^\circ$$

$$\therefore \theta = 0^\circ, 74^\circ, 360^\circ$$

Many left at $\theta = 360^\circ$

Preliminary (Year 11) Mathematics Extension I
Assessment Task 3 - 2018

Q. 3

a) $3x^3 - 4x + 6 = 0$
 $\alpha + \beta + \gamma = 0$
 $\alpha\beta + \beta\gamma + \gamma\alpha = -\frac{4}{3}$
 $\alpha\beta\gamma = -\frac{6}{3} = -2$

i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
 $= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{-\frac{4}{3}}{-2} = \frac{4}{3} \times \frac{1}{2} = \frac{2}{3}$

ii) $\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3$
 $= \alpha\beta\gamma(\alpha^2 + \beta^2 + \gamma^2)$
 $= \alpha\beta\gamma[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)]$
 $= -2[0^2 - 2 \times -\frac{4}{3}]$ Most students got this question correct, however small minority was confused about the sign of final product.
 $= -2 \times \frac{8}{3} = -\frac{16}{3}$ Provided $\cos 2\theta = -1$

b) $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$

R.H.S

$$= \frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \frac{1 - [2\cos^2 \theta - 1]}{1 + 2\cos^2 \theta - 1}$$

$$= \frac{1 - 2\cos^2 \theta + 1}{2\cos^2 \theta}$$

$$= \frac{2(1 - \cos^2 \theta)}{2\cos^2 \theta}$$
 Some students followed a long and complicated way to get to the final answer.

$$= \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$= \tan^2 \theta$$

$$= \text{R.H.S.}$$

Year 11 Assessment Task (2018) - III

Q.3

c) $P(x) = (x-3)(x-4) * Q(x) + Ax + B$ — (1)

Sub $x=3$ in (1)

$$P(3) = 3A + B$$

But $P(3) = 5$

$$5 = 3A + B \quad \text{--- (2)}$$

Sub $x=4$ in (1)

$$P(4) = 4A + B$$

But $P(4) = 9$

$$9 = 4A + B \quad \text{--- (3)}$$

$$(3) - (2)$$

$$4 = A$$

Sub in (2)

$$5 = 12 + B$$

$$B = -7$$

Some students did not get the signs of A and B correct.

Hence Remainder = $4x - 7$

d) $y = \left(\frac{P+Q}{2}\right)x - aPQ$ (Given)

i) It passes through $(2a, 3a)$

$$3a = \left(\frac{P+Q}{2}\right)2a - aPQ$$

$$3 = P + Q - PQ$$

$$PQ = P + Q - 3$$

Hence proved.

Most students got it correct.

Year 11 Assessment Task -(2018)-III

Q3.

d

(ii)

$$M \left[\frac{2ap + 2a\phi}{2}, \frac{a(p^2 + \phi^2)}{2} \right]$$

$$M \left[a(p + \phi), \frac{a}{2} [(p + \phi)^2 - 2p\phi] \right]$$

But $p + \phi = p\phi + 3$ (Proved ~~above~~ previously)
so

$$M \left[a(p\phi + 3), \frac{a}{2} [(p\phi + 3)^2 - 2p\phi] \right]$$

Hence proved

Some students did not use $(a+b)^2 - 2ab = a^2 + b^2$,
and ended up getting very complicated approach

(iii)

$$x = a(p\phi + 3)$$

$$p\phi + 3 = \frac{x}{a}$$

$$y = \frac{a}{2} [(p\phi + 3)^2 - 2p\phi]$$

so

$$y = \frac{a}{2} \left[\frac{x^2}{a^2} - 2 \left[\frac{x}{a} - 3 \right] \right]$$

$$y = \frac{a}{2} \left[\frac{x^2}{a^2} - \frac{2x}{a} + 6 \right]$$

$$y = \frac{x^2}{2a} - x + 3a$$

$$\text{Also } x^2 = 4ay$$

$$y = \frac{2ay}{2a} - x + 3a$$

$$y = 2y - x + 3a$$

$$y - x + 3a = 0$$

The most economical way is to get $p\phi + 3 = \frac{x}{a}$ and sub it in y -coordinate of the ~~mid~~ point M and then to establish the relation between x and y .

Question 4

$$\begin{aligned} (a) \quad \# \text{ of arrangements} &= \text{total} - \# \text{ where they are together} \\ &= 7! - 6! \times 2! \\ &= 3600 \end{aligned}$$

Comment: Some students did not realise that there were 8 people in total. Also, some students did not consider that the Morrisons can be arranged in $2!$ ways.

(b)(i) In $\triangle AQB$ and $\triangle PBA$:

$$(i) \quad \angle QAB = \angle BPA \quad (\angle \text{ in alt. segment})$$

$$(ii) \quad \angle AQB = \angle PBA \quad (\angle \text{ in alt. segment})$$

$$\therefore \triangle AQB \parallel \triangle PBA \quad (\text{equiangular})$$

$$(ii) \quad \frac{AB}{PA} = \frac{QB}{BA} \quad \left(\begin{array}{l} \text{corr. sides of similar } \Delta s \\ \text{in same ratio} \end{array} \right)$$

$$\therefore AB \times BA = PA \times QB \quad \text{i.e. } AB^2 = PA \times BQ$$

Comment: Students needed to identify correctly the two pairs of equal angles using correct geometrical reasoning. Some careless errors occurred in this question.

(c) $A(-3, 7)$ $B(1, a)$
 $k: 1$

$$\frac{k(1) - 3(7)}{k+1} = 0$$

$$k - 3 = 0$$

$$\therefore \underline{k = 3}$$

$$\text{and } \frac{k(a) + 1(7)}{k+1} = -2$$

$$\frac{3a + 7}{4} = -2$$

$$3a + 7 = -8$$

$$3a = -15$$

$$\therefore \underline{a = -3}$$

Comment: Many mistakes made with the ordering for the calculation.

(d) (i) $y = \frac{-x^2}{9a}$ $y' = -\frac{2x}{9a}$

at $A(3at, -at^2)$ $m_T = \frac{-2(3at)}{9a}$
 $= -\frac{2t}{3}$

$$\therefore m_N = \frac{3}{2t}$$

Normal: $y - (-at^2) = \frac{3}{2t} (x - 3at)$

$$2t(y + at^2) = 3(x - 3at)$$

$$2ty + 2at^3 = 3x - 9at$$

$$\therefore 3x - 2ty = 2at^3 + 9at$$

Comment: Algebraic errors were not uncommon.

(d)(ii)

when $x = -at$, $y = ?$

$$3(-at) - 2ty = 2at^3 + 9at$$

$$2ty = -12at - 2at^3$$

$$y = -6a - at^2$$

$$\therefore B = (-at, -6a - at^2)$$

(iii) $A(3at, -at^2)$ $B(-at, -6a - at^2)$
 $-2 : 3$ (external division)

$$C = \left(\frac{3(3at) - 2(-at)}{-2+3}, \frac{3(-at^2) - 2(-6a - at^2)}{-2+3} \right)$$

$$= (11at, -at^2 + 12a)$$

(iv) Using C from (iii) $x = 11at \Rightarrow t = \frac{x}{11a}$

$$\therefore y = -a \left(\frac{x}{11a} \right)^2 + 12a$$

$$= -\frac{x^2}{121a} + 12a \quad \therefore \frac{x^2}{121a} = -(y - 12a) \quad \leftarrow \text{directrix}$$

$$x^2 = -121a(y - 12a)$$



$\therefore$ A parabola with vertex at $(0, 12a)$.

$$4 \times \text{focal length} = 121a \quad \therefore \text{focal length} = \frac{121a}{4}$$

$$\text{Directrix is } y = 12a + \frac{121a}{4} \quad \text{i.e. } y = \frac{169a}{4}$$

Comment: (iii) Many mistakes due to ordering and/or calculating internal division instead of external.
(iv) Many students were unable to determine the equation of the directrix.